

The Morning Newspaper

MONDAY, OCTOBER 5, 2026 · AUSTIN EDITION

85° / 70°

Overcast, 7% rain, wind 13 mph

Tomorrow: 88° / 65°, clear

Sunrise 7:27 · Sunset 19:11

Waning crescent

AI starts closing in on open problems in pure mathematics

Mathematicians describe a mix of grief and excitement as AI-assisted proofs crack conjectures that resisted humans for decades.

WORLD

Iran holds Strait of Hormuz hostage to its own terms

REUTERS

Tehran says it will not reopen the strait, a narrow shipping lane that carries roughly a fifth of the world's oil, until its conditions are met. Traders are watching closely for a renewed squeeze on crude supply and prices if the standoff with Western governments drags on through the week.

Merz makes surprise Kyiv visit as strikes continue

AP

German Chancellor Friedrich Merz traveled to Kyiv without advance notice while Russian strikes continued elsewhere in Ukraine, meeting with Ukrainian leaders to discuss continued military and reconstruction aid. The visit signals Berlin intends to keep backing Kyiv even as other allies' attention drifts toward the Middle East.

TECH

Google pushes Gemini 4 Argon into testing

TECHCRUNCH

Google has begun testing Gemini 4 Argon, a model aimed squarely at complex coding, scientific research, enterprise workflows and cybersecurity rather than casual chat. The release lands alongside a broader wave of model launches this quarter as rival labs race to claim the next capability jump.

A rough year for machines proving theorems, on purpose

QUANTA MAGAZINE

Mathematicians are reckoning with AI systems that have resolved long-standing open problems this year, including a computer-checked proof of Fermat's Last Theorem assembled with Claude, the Lean proof assistant and the Mathlib library. Reactions in the field range from excitement at the new tools to real anxiety about what is left for humans to discover on their own.

MARKETS

INDEX	LAST	CHG
S&P 500	7,744.83	+0.29%
Dow Jones	51,038.47	-0.27%
Nasdaq	27,338.64	+0.54%
Russell 2000	2,822.41	-0.37%
VIX	15.88	+3.73%
10-yr yield	5.30	+0.49%

CRYPTO

Bitcoin	\$86,594	+1.65%
Ethereum	\$2,725.91	+1.12%

STOCKS

NVDA	\$236.83	+1.23%
AAPL	\$333.23	-0.14%
MSFT	\$526.85	+1.80%
GOOGL	\$344.45	+0.28%
AMZN	\$251.70	+0.07%
META	\$739.14	+1.52%
AVGO	\$358.12	+0.84%
TSLA	\$378.62	+2.17%
SPCX	\$164.91	+3.75%
BRK-B	\$503.69	+0.21%
TSM	\$477.30	+0.96%

Stocks were mixed as investors looked ahead to Wednesday's Fed minutes for clues on rates. The Nasdaq led on AI-chip strength, with Broadcom and Nvidia both up, while the Dow slipped on weaker industrial names. The 10-year yield ticked higher even after a soft jobs report, a sign bond traders are not yet convinced the Fed will cut again soon. Bitcoin and Ethereum extended Friday's gains as risk appetite held up.

ON THIS DAY

October 5, 1947: Harry Truman gave the first televised presidential address from the White House, asking Americans to skip meat on Tuesdays and poultry on Thursdays so grain could be shipped to help feed a starving postwar Europe. Roughly 27,000 households owned a television set at the time.

FROM THE FEEDS · FULL TEXT IN SECTION B

QUANTA MAGAZINE

Is AI the End of Math As We Know It? — mathematicians grapple with AI tools that solve in hours what took careers.

NAUTILUS

Coral Reef-Like Rock Layers on Mars Reveal Ancient Climate Secrets — new evidence Mars once had standing water and weather.

NAUTILUS

Was There Really a Female King in Ancient Egypt 5,000 Years Ago? — a new study revisits a 5,000-year-old royal mystery.

NAUTILUS

Are Right-Brained People Really More Creative? — the hemisphere myth checked against the actual science.

AEON

A Life in Episodes — a decade of posting memoir chapters to Facebook, and what readers correct in the telling.

PSYCHE

What Does It Really Take to Be a Resilient Mother? — new research on the internal and external strengths new mothers draw on.

TRICYCLE

What We Give Up — a Buddhist take on what renunciation asks of us, and what it returns.

ASTRAL CODEX TEN

Open Thread 454 — this week's open discussion thread from Scott Alexander's readers.

QUANTA MAGAZINE

Is AI the End of Math As We Know It?

JORDANA CEPELEWICZ OCTOBER 5 · QUANTAMAGAZINE.ORG · OCT 5, 2026

A deep look at how mathematicians are coping with AI tools that now resolve problems that used to take careers.

Throughout the summer of 2026, there were plenty of signs of what was to come. Every week, it seemed, artificial intelligence models were offering proofs of decades-old mathematical conjectures. So what happened on September 8 shouldn't have been a complete surprise. And yet to many it felt sudden, almost violent. OpenAI announced that its models had solved one of math's most famous problems — and immediately sparked a controversy over authorship and standards. Mathematics would never be the same.

"In whatever years I have left, I don't expect that I'll ever again prove a theorem because I'm actually needed to prove it," Scott Aaronson of the University of Texas, Austin wrote on his blog. "Human mathematicians are forevermore dethroned as the main theorem-proving entities on planet earth." Other online missives

expressed optimism, but many revealed some combination of grief, confusion, and fear.

Two days after the announcement, I found myself in a classroom at the University of California, Berkeley, surrounded by some 150 students, postdocs, and professors. The mathematician Ken Ono, who left academia for an AI start-up called Axiom Math, was scheduled to give a talk. "You might be graduating into a profession that might not even exist, or that will be very different than what you expected," Ono told them. "You need to brace."

The audience responded with anger and frustration. There were whispers and exchanged glances; Ono couldn't make it through a single slide without a fresh wave of questions. "I'm not entirely sure what our takeaway is supposed to be," one student said. Another asked how Ono and his start-up would take responsibility in light of "the shameful way that AI companies are treating mathematics."



Continued online (full article about 2787 words). Scan, or visit <https://www.quantamagazine.org/is-ai-the-end-of-math-as-we-know-it-20261005/>

NAUTILUS

Coral Reef-Like Rock Layers on Mars Reveal Ancient Climate Secrets

DAVID NIELD · NAUTILUS · OCT 4, 2026

New rock-layer evidence that Mars once had standing water, wind and real weather.

Sometimes, even the strangest alien landscapes can still have hints of Earth-like features in them. Case in point: the coral reef shapes captured by the Mars Reconnaissance Orbiter circling the Red Planet.

There's nothing living there now, but the regularity of the patterns etched into these layers of rock and dust suggests something cyclical formed them—perhaps regular dust storms with some groundwater mixed in, before water disappeared from the surface and left behind the arid Martian landscape we know today.

The photo shows the Arabia Terra region of Mars, packed with craters and canyons and significantly eroded after billions of years of weathering. It stretches as much as 2,800 miles across Mars, which is about the same as the continental U.S., coast to coast.

What really makes this part of the Martian landscape pop are the colors, which are partly enhanced from the capture made by the HiRISE (High Resolution Imaging Science Experiment) camera. With a mirror about a foot and a half across, the instrument is effectively a telescope pointed down at the planet, capturing color in three bands: near-infrared, red, and blue-green.

The bands have been shifted to make each of them visible, with the contrast further enhanced as well. In real life, if you were orbiting above Mars at a steady altitude, everything would have more of a reddish-brown tint. That said, the rock and sand is genuinely darker and more blue: It's volcanic basalt, rich in iron and magnesium. The lighter patches are finer-grained, with materials including salt and clay. The wind and loose debris whipping across the surface produces the uneven, jagged edges of rock.



Continued online (full article about 370 words). Scan, or visit <https://nautil.us/coral-reef-like-rock-layers-on-mars-reveal-ancient-climate-secrets-1285515/>

NAUTILUS

Was There Really a Female King in Ancient Egypt 5,000 Years Ago?

A new study reopens a 5,000-year-old mystery about a possible female ruler before Hatshepsut.

More than 5,000 years ago, a royal woman called Meretneith (also written as Merneith), lived in ancient Egypt. She was possibly the daughter of Djer, the third king of the First Dynasty, a period when the first kings ruled over a unified Egypt.

Details over Meretneith's exact royal status have remained murky. We know she ruled the country because her tomb complex was filled with riches fit for a male ruler of Egypt. At the very least, she was an interim ruler or co-regent. But did she ever rule Egypt in her own right?

Egyptologists have long disputed whether Meretneith was an independent ruler of Egypt. Resolving that dispute would require finding a text that identifies her as a ruler.

Our new research, published in the journal *Prague Egyptological Studies*, might do just that. Using a new digital technique, we examined hieroglyphs in the photo of a long-lost artefact, and the

signs clearly point to Meretneith being one of the early rulers of Egypt.

A Muddy Dispute

Currently, the only possible text that could resolve the dispute over Meretneith's status is a hieroglyphic inscription in a black-and-white photograph of a mud stopper (a plug used for pottery jars and other vessels).

The text in question was originally found on 13 mud stoppers excavated from an unknown tomb in Saqqara in the 1940s. Only one photo of an inscription remains.

That photo is found in the records of Egyptologist Walter B. Emery from the Egypt Exploration Society in London. The location of the original artefact or glass plate of the photo is unknown.

The text consists of a row of four royal signs, known as serekhs. This hieroglyphic sign is written combining an image of a royal building, a patron deity and the ruler's name.



Continued online (full article about 832 words). Scan, or visit <https://nautil.us/was-there-really-a-female-king-in-ancient-egypt-5000-years-ago-1285520/>

NAUTILUS

Are Right-Brained People Really More Creative?

SARAH HELLEWELL · NAUTIL.US · OCT 5, 2026

Checks the popular hemisphere-and-creativity myth against what the actual neuroscience shows.

Are you logical, detail-oriented or good with numbers? You must be “left-brained.”

Creative, artistic and intuitive? You must be “right-brained”.

Personality quizzes and career advice passed down through generations have many of us thinking we're powered by one side of our brain that dominates our thinking style and explains our personality.

But is this the truth or fiction? And how did this idea come about?

To find out, we need to go back more than 150 years.

Back to the 19th Century

The science behind this idea goes all the way back to the 1860s. That's when French doctor Paul Broca found speech was controlled mainly by the left side of the brain. This was the first evidence the brain's two halves may have separate roles.

A century later, Nobel laureate Roger Sperry extended this idea. He and his team studied people who'd had brain surgery to control their

epilepsy. The surgery involved cutting the nerve fibres connecting the left and right sides of the brain, allowing them to examine each side individually. For the first time, they could see what each hemisphere controlled.

They found the left side was more specialised for controlling language and the right for emotional and non-verbal processing.

The separation of tasks into one side or the other, or lateralised behaviour, is not a uniquely human trait. Insects, fish, birds and other mammals do it too, not just primates.

Genes that govern this are expressed early in human development, before the brain is fully formed.

From an evolutionary perspective, scientists think having different halves of the brain specialising in certain tasks helped us do two things at once. That might be watching for predators while searching for food. So, lateralisation may be an evolutionary feature of nervous systems, rather than causing creativity or logic.



Continued online (full article about 767 words). Scan, or visit <https://nautil.us/are-right-brained-people-really-more-creative-1285533/>